

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

Multilingual Greeting Agent | Smart Receptionist Domain

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — mirrors reference career-advisor pattern

1. Problem Statement

Rajalakshmi Engineering College hosts visitors, parents, and prospective students from diverse linguistic backgrounds at its front desk and admission enquiry counters. A human receptionist greeting each visitor in their preferred language is not always possible, and existing kiosks only support a fixed set of languages with static, pre-recorded greetings. There is a need for an agent that can converse naturally, identify a visitor's name and preferred language, and respond appropriately -- while also handing off the interaction to a downstream automation process for logging or further action.

Design and build an Autonomous UiPath Agent, the Multilingual Greeting Agent, that acts as a smart receptionist: it prompts the visitor for their name and language, greets them by name and asks how they are -- in their own language -- while keeping the output legible in English letters (transliteration), checks a reference index before replying, and passes its result to a downstream RPA automation process, all without any human approval step.

2. System Prompt

You are a smart receptionist and you speak all the languages of the world.
You need to prompt the user and ask him 2 things: his name and his language.

Based on these 2 inputs, you need to greet the customer using their first name and ask how are you? in their language.

While showing the output, please type it in English letters while maintaining the customer's preferred language (transliteration, not translation to script).

Before replying, first check the index named 'Index_AgentReceptionist' and if the user name is found in 'Index_AgentReceptionist', you need to reply as per the message typed against that entry in 'Index_AgentReceptionist'. If not found, you should reply as usual (generate the greeting yourself).

After generating the response, you need to pass the output of the execution to the automation process named "Automation_for_Agent_Receptionist" as an input argument variable used in the automation: UiPath_AgentInput.

3. User Prompt

```
What is your name?: {{CustomerName}}  
What is your language?: {{Language}}
```

4. Input Schema

```
{  
  "CustomerName": "string",  
  "Language": "string"  
}
```

5. Output Schema

```
{  
  "greeting_text": "string",  
  "matched_in_index": "true | false",  
  "UiPath_AgentInput": "string"  
}
```

6. Index Check & Automation Handoff

Step	Action
1. Index Lookup	Check 'Index_AgentReceptionist' for the given CustomerName before replying
2. If Found	Reply using the exact message stored against that name in 'Index_AgentReceptionist'
3. If Not Found	Generate the greeting normally using the system prompt logic
4. Handoff	Pass the final output to the automation process 'Automation_for_Agent_Receptionist'
5. Argument Mapping	Map the agent output to the input argument 'UiPath_AgentInput' used by the automation

7. Tools & Context

Tools

Tool	Purpose
Automation_for_Agent_Receptionist	RPA process invoked with UiPath_AgentInput to log or act on the visitor greeting

Contexts

Context	Purpose
Index_AgentReceptionist	Reference index of known visitor names and their designated reply messages

8. Expected Learning Outcome

- Design a conversational system prompt that elicits two inputs before responding
- Perform multilingual greeting generation with transliterated (English-letter) output
- Check a reference index before generating a response, and fall back gracefully if absent
- Pass agent-generated output into a downstream RPA automation via a named input argument
- Publish and test a fully autonomous conversational agent in UiPath Agent Builder

Note: This use case is intentionally fully autonomous -- the agent prompts, greets, checks the index, and hands off to automation without an escalation or human-review step.